

Course: AI and Cybersecurity Applications (PGR216) (Bachelor level)

Exam Type: Home exam

Duration: 1 Week

Allowed Resources: All (open book, open internet)

Submission: Report (PDF), 20 pages max

Submission language: The report may be written in either English or Norwegian. Students are nevertheless encouraged to write in English, since the lectures and course materials are provided in English.

1. General Description:

This home exam assesses your ability to reason critically, design AI solutions, and reflect on their applicability in real world scenarios. The exam emphasizes logical thinking, experimental design, and interpretation rather than achieving maximum model accuracy. You are expected to make independent decisions, justify them clearly, and demonstrate understanding of both AI methods and the real-world challenges.

2. Technical Requirements (Python Environment):

You may implement all models and experiments in Python.

Required Core Packages

You are expected to use standard, well-established libraries:

- Python ≥ 3.9
- NumPy
- Pandas
- SciPy
- scikit-learn
- Matplotlib
- Seaborn

Deep Learning Libraries

(Relevant for question 4 only)

- PyTorch (+ torchvision, torchmetrics)

Data Handling & Utilities

- Jupyter Notebook or equivalent
- tqdm (optional, progress tracking)
- imbalanced-learn (if handling class imbalance)

Explicit Restrictions

- AutoML tools (e.g., AutoKeras, H2O, AutoGluon): NOT allowed
- Copying full pipelines from blogs/tutorials without modification: NOT allowed

3. Experimental Rules

To ensure originality and reduce cheating:

- You must justify:
 - Feature selection
 - Model architecture
 - Evaluation metrics
- Performance numbers alone are not sufficient

- You must include:
 - What did not work
 - What assumptions your model makes
 - Where your model would fail in real deployment

Identical or near-identical experimental setups across submissions may result suspicious of cheating and may result lower grade.

4. Report Requirements (Strict)

- Maximum: 20 pages (Font: Calibri, Times New Roman, or Ariel with font size of 11)
- References are excluded from the page limit
- Appendices are optional.
- Submissions exceeding 20 pages may lower the grade.

5. Tentative Report Structure

Each question could be answered using the following template.

- **Introduction**
 - Problem motivation
- **Dataset Understanding**
 - Dataset characteristics, features and attributes
 - Limitations and biases in the dataset if any
- **Methodology**
 - Model choice and justification
 - Experimental design
- **Results**
 - Quantitative results
 - Clear explanation of metrics
 - Comparison where applicable
 - Visualizations

Learning curves

- Confusion matrices
 - Error analysis plots
- (Figures must support reasoning, not decoration)

- **Personal Reflection (Mandatory)**
 - What assumptions your models rely on
 - What surprised you
 - What you would change in a real system
- **Discussion**
 - Strengths and weaknesses
 - Deployment considerations
 - Ethical and operational risks

6. Code Submission Requirements

- Only report will be graded. Code submission is optional.
- If code is submitted with report, it should be reproducible and include:
 - requirements.txt or environment description
 - Clear instructions to run experiments

- You may submit A ZIP file, or a private Git repository (access provided to instructors)

7. Academic Integrity

This is an individual exam.

You may:

- Use documentation
- Use general examples
- Discuss concepts at a high level

You may not:

- Copy solution structures from others
- Copy online blogs/tutorials with any changes and personal reasoning.
- Submit AI-generated text without personal reasoning

All submissions must reflect your own thinking.

Exam Questions:

Question 1: (20% weightage)

Problem Statement

The UCI Energy Efficiency dataset contains building-design samples described by 8 input variables and 2 continuous target variables: Heating Load (Y1) and Cooling Load (Y2). Each row represents one building design configuration. Your task is to analyze the dataset and develop a well-reasoned regression pipeline to study how the input features influence the two energy-load targets.

Your analysis should not only focus on prediction performance, but also on understanding the dataset, justifying preprocessing decisions, interpreting relationships between variables, and reflecting on model behavior. In particular, you should determine whether Heating Load or Cooling Load is harder to predict and support your conclusion using evidence from your analysis.

Required Steps

- Perform exploratory data analysis (EDA) to understand the dataset structure, variables, and general data characteristics.
- Use suitable visualization techniques to explore the distributions of features and targets, and to interpret important relationships in the data.
- Conduct appropriate preprocessing, including discussion of feature handling and feature scaling, where relevant.
- Perform correlation analysis to examine relationships among the input variables and between the inputs and the two targets.
- Build and evaluate regression models for predicting Heating Load and Cooling Load.
- Compare model performance using appropriate evaluation and benchmarking measures.
- Discuss whether the problem should be treated as two separate regression tasks or as a multi-output regression task and justify your choice.

Mandatory Reflection

- Based on your exploratory analysis, preprocessing, correlation analysis, and regression results, discuss whether Heating Load or Cooling Load is harder to predict.
- Your answer must be evidence-based and should connect observed model performance with patterns found in the dataset.
- Reflect on which preprocessing or analysis steps were most important for achieving good performance and meaningful interpretation.

Dataset

UCI Energy Efficiency Dataset

Link: <https://archive.ics.uci.edu/dataset/242/energy+efficiency>

Question 2: (20% weightage)

Problem Statement

You are given the UNSW-NB15 dataset for network intrusion detection. The dataset contains 49 extracted traffic features and supports classification across normal traffic and nine attack categories (1+9=10 classes). The dataset includes a mixture of numeric, binary, and nominal variables.

Your task is to develop and justify a complete multi-class classification pipeline. In addition to building predictive models, you must examine the dataset carefully, handle encoded features appropriately,

address class imbalance, and evaluate performance using metrics that reflect the difficulty of minority class prediction.

Required Steps

- Carry out EDA to understand the structure and quality of the dataset.
- Identify the target classes and analyze the class distribution.
- Determine which features require encoding, apply an appropriate encoding strategy, and justify your choice.
- Apply relevant preprocessing and feature handling.
- Train a baseline multi-class classification model.
- Apply one or more techniques to reduce the effect of class imbalance and compare the results against the baseline.
- Evaluate model performance using appropriate multi-class benchmarking metrics, not only overall accuracy.
- Analyze per-class performance and discuss which classes are more difficult to classify.

Mandatory Reflection

Using evidence from your experiments, discuss how encoding choices and class imbalance handling influenced the final results. Your answer should also explain why strong overall accuracy may still hide weak performance on minority attack classes.

Dataset

UNSW-NB15 Dataset

Link: <https://www.kaggle.com/datasets/dhoogla/unswnb15>

Question 3: (30% weightage)

Problem Statement

This question extends the analysis from Question 1 and Question 2 by focusing on the bias-variance trade-off and its impact on model generalization. You are required to examine how regularization, feature dimensionality reduction, and cross-validation affect the performance and stability of your models.

Required Steps

- Explain the concepts of bias, variance, underfitting, and overfitting.
- Analyze whether your initial models in Question 1 and Question 2 show signs of high bias or high variance.
- Apply and evaluate regularization methods.
- Apply and evaluate dimensionality reduction of features.
- Use cross-validation to compare models more reliably.
- Discuss how these techniques influence generalization and model stability.

Mandatory Reflection

Using evidence from your experiments, discuss whether the main challenge in each question was bias or variance, and explain how regularization, dimensionality reduction, and cross-validation helped you better understand and improve the models.

Question 4: (30% weightage)

Problem Statement

The Maling dataset is a benchmark dataset for image-based malware classification. Each sample is a grayscale image generated from a malware binary and labeled with its corresponding malware family.

Your task is to design and justify a complete image-classification pipeline for malware family prediction. In addition to model performance, you must examine preprocessing choices, class imbalance, and class-wise behavior.

Required Steps

- Perform EDA to understand the dataset structure, image properties, and class distribution.
- Visualize samples from different malware families and comment on visible similarities and differences.
- Apply suitable image preprocessing, such as resizing and normalization.
- Either design a CNN model or apply transfer learning for multi-class malware family classification.
- Since the dataset is imbalanced, apply one or more methods to reduce the effect of class imbalance and compare the results with the baseline model.
- Evaluate the models using appropriate multi-class classification metrics, not only overall accuracy.
- Analyze per-class performance and discuss which malware families are easier or harder to classify.
- Examine common misclassification patterns and discuss whether some malware families appear visually similar.

Mandatory Reflection

Using evidence from your experiments, discuss how class imbalance affected model performance and whether the CNN appears to learn meaningful malware structure or mainly superficial visual patterns.

Dataset

Malimg Dataset

Link: <https://www.kaggle.com/datasets/manmandes/malimg>